SOMAIYA

VIDYAVIHAR

K J Somaiya Institute of Technology

An Autonomous Institute Permanently Affiliated to the University of Mumbai

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Class : Second Year (SY) 2026-27 **Department : Information Technology**

Course code: ITXS37 **Subject: DSA**

Experiment No:1

Aim: Experiment No. 01 (Implementation of Stack using Array)

Program:

```
#include <stdio.h>
#include <stdlib.h>
#define MAX 1;
int stack[MAX];
int top = -1;
int isFull() {
    return top == MAX - 1;
}
int isEmpty() {
    return top == -1;
}
void push(int temp) {
   if (isFull()) {
  printf("Stack Overflow! Cannot push more alerts.\n");
```

```
        return;
    }
    stack[++top] = temp;
    printf("Pushed temperature %dC \n", temp);
}
int pop() {
    if (isEmpty()) {
        printf("Stack is empty! \n");
        return -1;
    }
    return stack[top--];
}
void clearAllAlerts() {
    if (isEmpty()) {
        printf("No active alerts to clear.\n");
        return;
    }
    printf("Temperature normalized :\n");
    while (!isEmpty()) {
        int temp = pop();
        printf("  %dC\n", temp);
    }
    printf("Stack is now empty.\n");
}
```

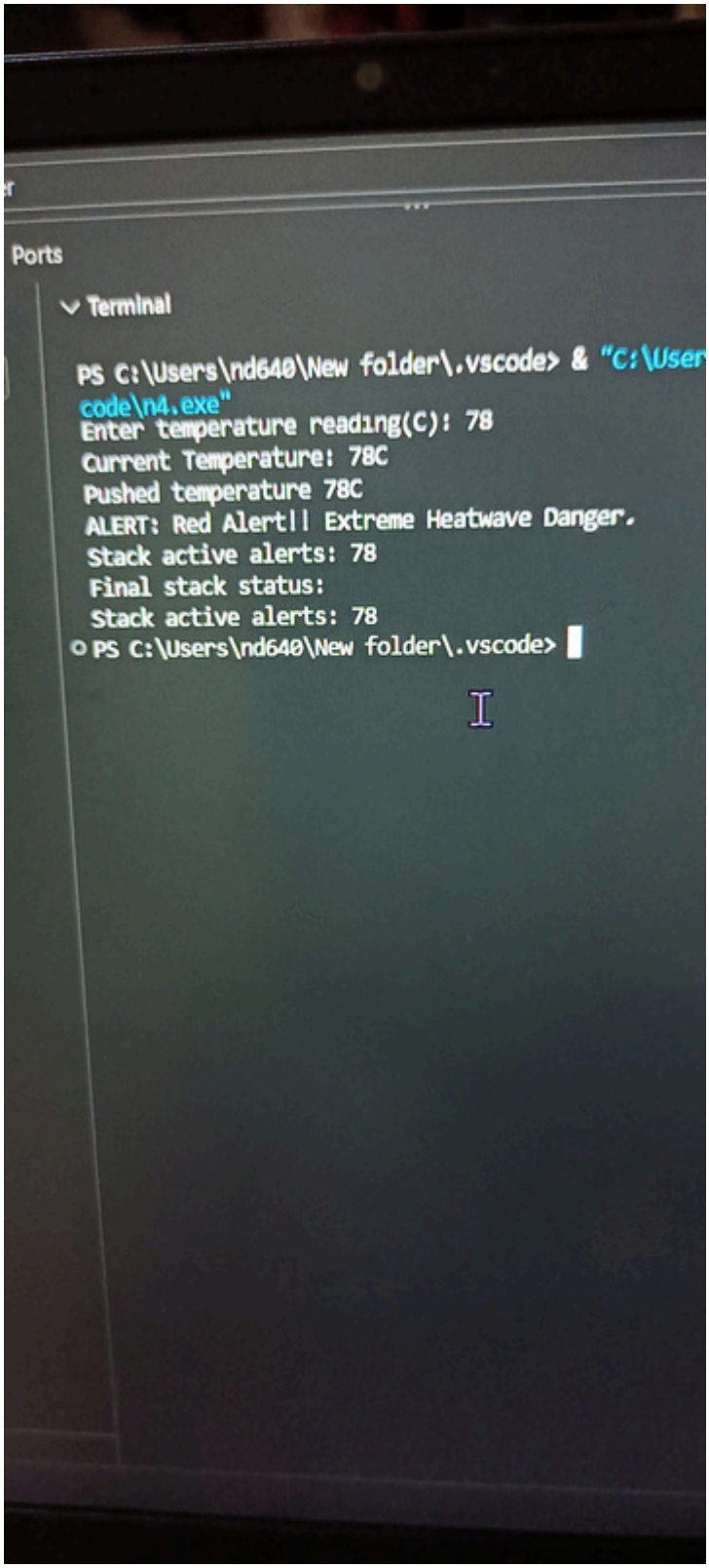
```
void display() {
    if (isEmpty()) {
        printf("Stack : [ EMPTY ]\n");
        return;
    }
    printf("Stack active alerts: ");
    for (int i = top; i >= 0; i--) {
        printf("%d ", stack[i]);
    }
    printf("\n");
}
void checkTemperature(int temp) {
    printf("Current Temperature: %dC\n", temp);
  if (temp >= 45) {
   push(temp);
        printf("ALERT: Red Alert!! Extreme Heatwave Danger.\n");
   }
    else if (temp >= 43) {
        push(temp);
        printf("ALERT: Orange Alert! Severe Heatwave Warning.\n");
    }
    else if (temp >= 40) {
        push(temp);
        printf("ALERT: Yellow Alert. Heatwave Watch.\n");
```

```
    }
    else {
        clearAllAlerts();
    }
    display();
}
int main() {
    int n, temp;
   printf("Heatwave Alert Prediction System \n");
    printf("Enter number of temperature readings to process: ");
  scanf("%d", &n);
    for (int i = 0; i < n; i++) {
        printf("\nEnter temperature reading(C): ");
        scanf("%d", &temp);
        checkTemperature(temp);
   }
    printf("Final stack status:\n");
    display();
    return 0;
}
```

Outcome:

```
PS C:\Users\nd640\New folder\.vscode> & "C:\User
code\n4.exe"
Enter temperature reading(C): 78
Current Temperature: 78C
Pushed temperature 78C
ALERT: Red Alert!! Extreme Heatwave Danger.
Stack active alerts: 78
Final stack status:
Stack active alerts: 78
PS C:\Users\nd640\New folder\.vscode>
```

Conclusion: SUCESSFULLY EXCUTED THE STACK USING STRING


Ports
Terminal
PS C:\Users\nd640\New folder\.vscode> & "C:\User
code\n4.exe"
Enter temperature reading(C): 78
Current Temperature: 78C
Pushed temperature 78C
ALERT: Red Alert|| Extreme Heatwave Danger.
Stack active alerts: 78
Final stack status:
Stack active alerts: 78
PS C:\Users\nd640\New folder\.vscode>